Experiment Set 1

Submission Due Date: 21/08/2026

References and Resources

(1) Website: https://www.arduino.cc/

(2) Arduino IDE Environment with libraries

Experiment Set-up Requirements

- Arduino UNO board
- Arduino IDE (Download from www.arduino.cc)
- Additional Components required:
  - Bread board
  - Resisters
  - Potentiometer (Variable Resister)
  - LEDs
  - Light Dependent Resistor (LDR)
  - Push Button Switch

Initialization

After connecting Arduino UNO to IDE, use IDE menu for the following initializations.

1. Tools > Board > Arduino AVR Boards > Arduino Uno

2. Tools > Port > COM3 (Arduino UNO)

Experiment 1

1. Use Program Titled “Experiment_1”
2. Observe LED on Arduino Board
3. Connect external LED on Pin 13, use current limiting resister, observe the blinking of LED.
4. Limit current through externally connected LED to 20mA. So, use appropriate resistor.
5. Understand the complete program written on the next page.

```
void setup() {
  // initialize digital pin 13 as an output.
  pinMode(13, OUTPUT);
}

// the loop function runs over and over again forever
void loop() {
  digitalWrite(13, HIGH);   // turn the LED on (HIGH is the voltage level)
  delay(1000);              // wait for a second
  digitalWrite(13, LOW);    // turn the LED off by making the voltage LOW
  delay(1000);              // wait for a second
}
```

Answer Following Questions in your report..

Question 1.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 1.b: Describe and explain following functions with one example.

(1.b.1) Setup

(1.b.2) Loop

(1.b.3) pinMode

(1.b.4) digitalWrite

(1.b.5) delay

Question 1.c: To limit the current through externally connected LED to 20mA, calculate the minimum value of series resistor that you may want to use.

Answer Following Questions in your report..

Question 1.d: What is DC current rating per i/o pin ?

Question 1.e: What is DC current rating for 3.3V pin?

Question 1.f: What is total number of digital i/o pins?

Question 1.g: What is total number of analog i/p pins?

Question 1.h: How many digital i/o pins have PWM facility?

Question 1.i: What is EEPROM capacity?

Question 1.j: What is SRAM capacity?

Question 1.k: What is Flash Memory capacity?

Question 1.l: What can be the maximum size of the program/code that you can write into the Arduino UNO?

Question 1.m: Write at-least one advantage of Arduino Mega over Arduino UNO.

Question 1.n: Write at-least one advantage of Arduino Nano over Arduino UNO.

Question 1.o: Assume that you write one program/code and then execute that code on three different Arduinos – Mega, UNO and Nano. Please comment on the comparative speed of the execution (time taken to execute that program) on all the three mentioned Arduinos. Explain the criteria that determine the speed of the execution of that code.

Experiment 2

Modify the program in Experiment No. 1 to generate 40% duty cycle for LED to remain turn ON.

That means, the LED will be OFF for 60% of time and ON for 40% of time.

Answer Following Questions in your report..

Question 2.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 2.b: Include your new c program in your report.

Question 2.c: Describe the changes you have made in the program.

Question 2.d: Show your calculations that demonstrate the 40% duty cycle.

Experiment 3

Modify and update hardware circuit, without changing the program of Experiment No. 2, such that two LEDs behave in following way.

LED 1: will be OFF for 60% of time and ON for 40% of time.

LED 2: will be OFF for 40% of time and ON for 60% of time.

Answer Following Questions in your report..

Question 3.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 3.b: Describe the changes you have made in the hardware circuit connections.

Experiment 4

1. Use program “Experiment_4”
2. Connect external LED on Pin 9, use current limiting resister, observe the fading of LED light.
3. Understand the complete program written on the next page.

```
int led = 9;           // the PWM pin the LED is attached to
int brightness = 0;    // how bright the LED is
int fadeAmount = 5;    // how many points to fade the LED by

// the setup routine runs once when you press reset:
void setup() {
  // declare pin 9 to be an output:
  pinMode(led, OUTPUT);
}

// the loop routine runs over and over again forever:
void loop() {
  // set the brightness of pin 9:
  analogWrite(led, brightness);

  // change the brightness for next time through the loop:
  brightness = brightness + fadeAmount;

  // reverse the direction of the fading at the ends of the fade:
  if (brightness <= 0 || brightness >= 255) {
    fadeAmount = -fadeAmount;
  }
  // wait for 30 milliseconds to see the dimming effect
  delay(30);
}
```

Answer Following Questions in your report..

Question 4.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 4.b: Explain following function.

- ✓ analogWrite

Question 4.c:

Explain following code:

```
if (brightness <= 0 || brightness >= 255) {
    fadeAmount = -fadeAmount;
  }
```

Question 4.d: On which output pins of Arduino UNO can we run program for fading LED? List the number of pins and Explain the reason.

Experiment 5

Write a program to blink LED on Pin 13 and Fade the LED on Pin 9 simultaneously.

Connect the hardware and observe the output of the program.

Save this new program.

Answer Following Questions in your report..

Question 5.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 5.b: Include your new c program in your report.

Question 5.c: Describe the changes you have made in the program and the circuit connections.

Experiment 6

Write a program to blink LED on Pins 13 , 11, 9 and 7 simultaneously every second.

Execute the program and observe the output on hardware.

Save this new program.

Answer Following Questions in your report..

Question 6.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 6.b: Include your new c program in your report.

Question 6.c: Describe the changes you have made in the program and the circuit connections.

Experiment 7

Write a program -

(1) to turn on LED on Pins 13 , 11, 9 and 7 one after another with delay of 2 second. (This means, first LED on pin 13 turns on. After 2 second delay, second LED on pin 11 turns on. This continues till all four LEDs turn ON)

(2) After all four LEDs are on, turn all of them OFF together.

(3) Repeat above steps continuously.

Execute the program and observe the output on hardware.

Save this new program.

Answer Following Questions in your report..

Question 7.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 7.b: Include your new c program in your report.

Question 7.c: Describe the changes you have made in the program and the circuit connections.

Simplified Traffic Signals

Experiment 8

Assume four LEDs are connected to Pins 13, 11, 9 and 7 of Arduino UNO. Write a program to turn on one LED at a time.

Second 1: LED on Pin 13 turns on, all other LEDs are off
Second 2: LED on Pin 13 turns off, LED on pin 11 turns on, other LEDs are off.
Second 3: LED on Pin 11 turns off, LED on pin 9 turns on, other LEDs are off
Second 4: LED on Pin 9 turns off, LED on pin 7 turns on, other LEDs are off.
Second 5: LED on Pin 7 turns off, LED on pin 13 turns on, other LEDs are off.

Use delay of 2 Seconds between each transactions.

Above cycle repeats continuously.

Execute the program and observe the output on hardware.

Save this new program.

Answer Following Questions in your report..

Question 8.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 8.b: Include your new c program in your report.

Question 8.c: Describe the changes you have made in the program and the circuit connections.

Answer Following Questions in your report..

For the following questions, refer the Datasheet of LED shared in the folder.

Question 8.d: Which material combination is used to generate Green colour ?

Question 8.e: What is peak forward current for Red LED ?

Question 8.f: What is peak wavelength for Blue LED?

General Questions:

Question 8.g: Why we need different materials (and combinations of different materials) to generate different colors of LED light? Explain with diagram.

Question 8.h: What is “Luminosity”? What is its SI unit?

Experiment 9

1. Use the program “Experiment_9”
2. Connect external LED on Pin 13, use current limiting resister.
3. Connect push button switch on Pin 2 of Arduino UNO. Connect a register in series with switch.

   Question: How will you generate two states (Logic low and logic high) using push button switch? Discuss the circuit with the instructor before implementing it.)
4. Run program mentioned in step 1 and observe that when switch is pressed, the LED turns off.

```
// constants won't change. They're used here to
// set pin numbers:
const int buttonPin = 2;     // the number of the pushbutton pin
const int ledPin =  13;      // the number of the LED pin

// variables will change:
int buttonState = 0;         // variable for reading the pushbutton status

void setup() {
  // initialize the LED pin as an output:
  pinMode(ledPin, OUTPUT);
  // initialize the pushbutton pin as an input:
  pinMode(buttonPin, INPUT);
}

void loop() {
  // read the state of the pushbutton value:
  buttonState = digitalRead(buttonPin);

  // check if the pushbutton is pressed.
  // if it is, the buttonState is HIGH:
  if (buttonState == HIGH) {
    // turn LED on:
    digitalWrite(ledPin, HIGH);
  } else {
    // turn LED off:
    digitalWrite(ledPin, LOW);
  }
}
```

Observation

Observe that when the switch is pressed, the LED turns off.

Understand the program written on the next page.

Answer Following Questions in your report..

Question 9.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 9.b: Explain following function.

- ✓ digitalRead

Question 9.c: Draw the circuit with the press button switch used for the input to the Arduino. Explain how this circuit will generate two logic states of “logic high” and “logic low”.

Experiment 10

Modify the program of Experiment 9 such that the LED turns OFF when the switch is pressed. If the switch is not pressed, then the LED should remain ON.

Save the new program.

Answer Following Questions in your report..

Question 10.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 10.b: Include your new c program in your report.

Question 10.c: Describe the changes you have made in the program and the circuit connections.